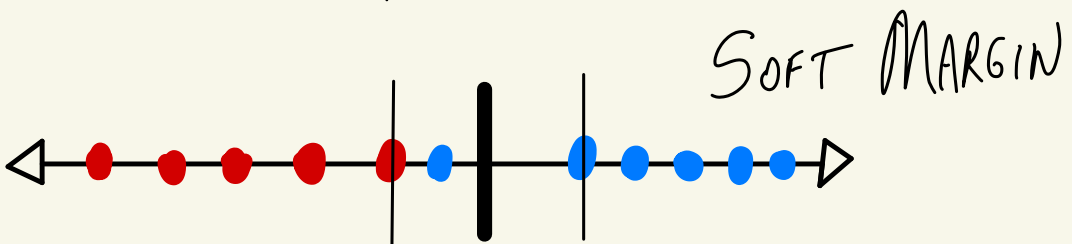
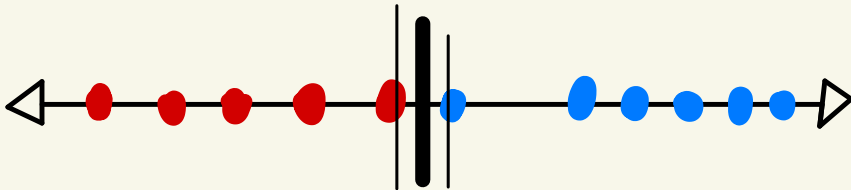
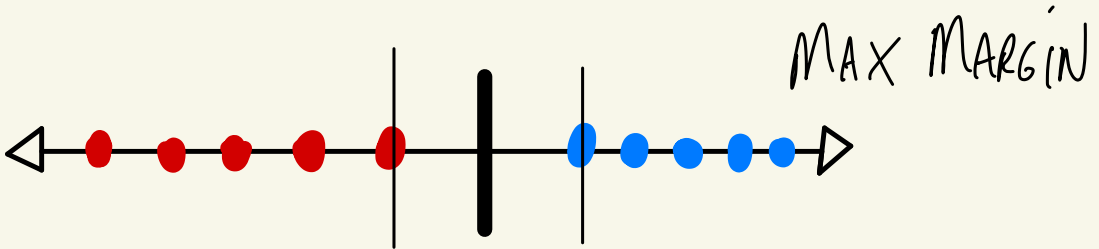
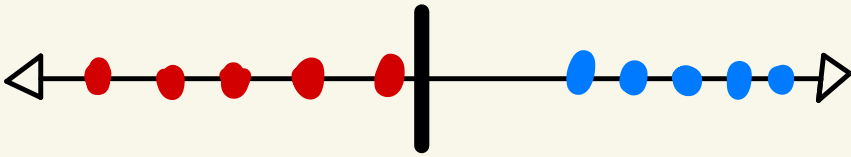


SVM CONCEPTUALIZATION

- 1) MOVE TO A HIGHER DIMENSION
KERNEL FUNCTION
- 2) FIND THE SUPPORT VECTOR
CLASSIFIER
 - MAX MARGIN
 - SOFT MARGIN
 - C PARAMETER

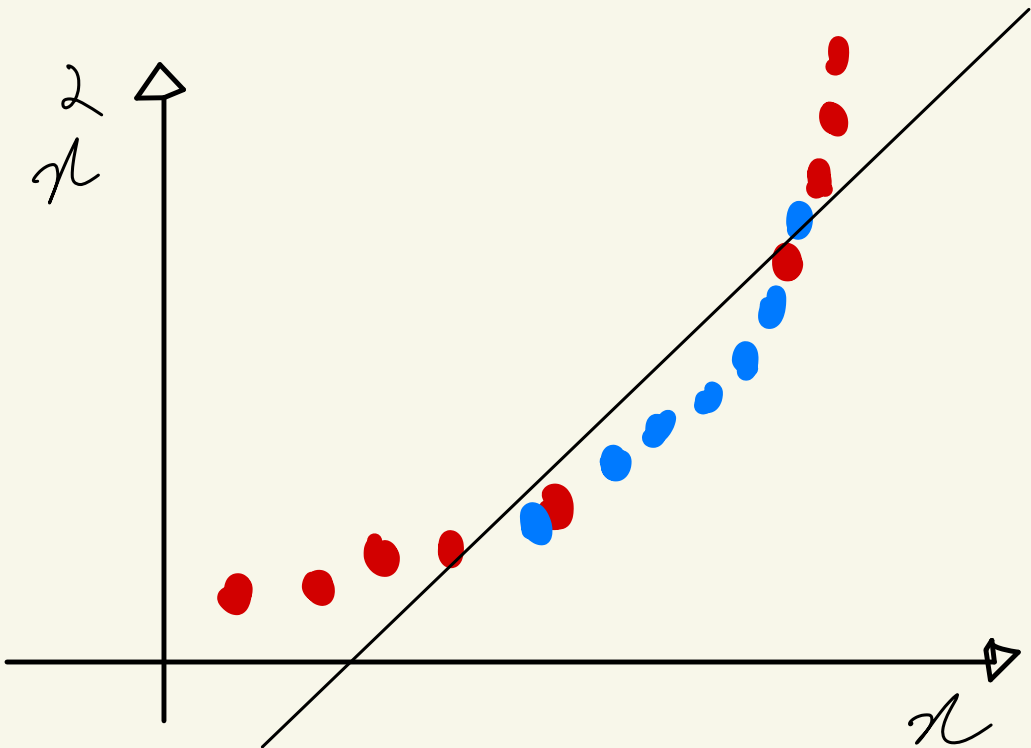
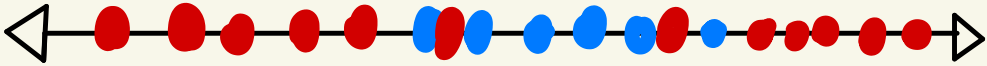
SVM CONCEPTUALIZATION

1D CASE SUPPORT VECTOR CLASSIFIER



SVM CONCEPTUALIZATION

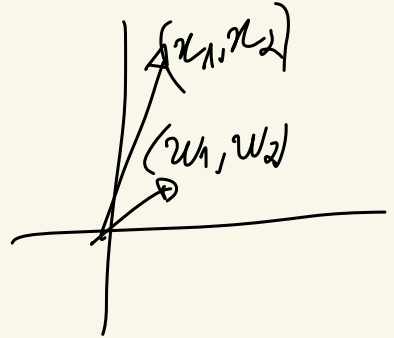
MOVE TO A HIGHER DIMENSION



SVM CONCEPTUALIZATION

$$\frac{w \cdot x + b}{\sim}$$

DOT PRODUCT



$$w_1 \cdot x_1 + w_2 \cdot x_2 + b$$

$\swarrow \quad \searrow$

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2$$

$$M = \frac{2}{|w|}$$

$$\begin{aligned} w \cdot x_1 + b &= +1 \\ w \cdot x_2 + b &= -1 \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \begin{aligned} w x_1 - w x_2 &= 2 \\ w (x_1 - x_2) &= 2 \end{aligned}$$

C PARAMETER

$$\max\left(\frac{2}{|w|}\right) \rightarrow \min\left(\frac{1}{2} |w|^2\right)$$

C PARAMETER ACTS LIKE A REGULARIZATION KNOB.

$$\min \frac{1}{2} |w|^2 + C \sum_{i=1}^m \zeta_i$$

PENALIZE
FOR MISCLASSIFICATION

POLYNOMIAL KERNEL

$$\begin{aligned}(a \cdot b + r)^2 &= a^2 b^2 + 2rab + r^2 \\ &= (a^2, \sqrt{r} \cdot a, r) \cdot (b^2, \sqrt{r} \cdot b, r)\end{aligned}$$

SUPPORT VECTOR MACHINES

- CONCEPTUALIZATION
- MAX MARGIN
- C PARAMETER FOR REGULARIZATION
- KERNEL FUNCTION & TRICKS
 - POLYNOMIAL KERNEL
 - SIGMA PARAMETER
- TIME COMPLEXITY